

9.8.19

BALU-ai25btech11017

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Question

- ① Two circles $x^2 + y^2 = 6$ and $x^2 + y^2 - 6x + 8 = 0$ are given. Then the equation of the circle through their points of intersection and the point $(1, 1)$ is (1980)
- ① $x^2 + y^2 - 6x + 4 = 0$
 - ② $x^2 + y^2 - 3x + 1 = 0$
 - ③ $x^2 + y^2 - 4y + 2 = 0$
 - ④ none of these

Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,
equation circles are

$$\mathbf{x}^T \mathbf{x} - 6 = 0 \quad (1)$$

$$\mathbf{x}^T \mathbf{x} + 2 \begin{pmatrix} -3 & 0 \end{pmatrix} \mathbf{x} + 8 = 0 \quad (2)$$

substitute 1.1 in 1.2

$$\begin{pmatrix} -3 & 0 \end{pmatrix} \mathbf{x} = -7 \quad (3)$$

$$x = \frac{7}{3} \quad (4)$$

substitute 1.4 in 1.1

$$y = \frac{\sqrt{5}}{3}, \frac{-\sqrt{5}}{3} \quad (5)$$

let equation of required circle is

Theoretical Solution

$$\mathbf{X}^T \mathbf{X} + 2\mathbf{u}^T \mathbf{X} + f = 0 \quad (6)$$

(7)

$$\begin{pmatrix} 2 & 14 & 14 \\ 2 & \frac{2\sqrt{5}}{3} & \frac{-2\sqrt{5}}{3} \\ 1 & 1 & 1 \end{pmatrix}^T \begin{pmatrix} \mathbf{u} \\ f \end{pmatrix} = \begin{pmatrix} -2 \\ -6 \\ -6 \end{pmatrix} \quad (8)$$

Theoretical Solution

$$f = 1 \quad (9)$$

$$\mathbf{u} = \begin{pmatrix} \frac{-3}{2} \\ 0 \end{pmatrix} \quad (10)$$

the required circle equation is

$$x^2 + y^2 - 3x + 1 = 0 \quad (11)$$

Plot

